

# SPACE312 Final Project

## GTO-to-GEO Transfer Trajectory Optimization

Mission target: GEO-KOMPSAT-2A

In this project, each team designs an optimized transfer trajectory from a geostationary transfer orbit (GTO) to the GEO location of GEO-KOMPSAT-2A. The mission must be solved as a trajectory optimization problem. The main trade space is total impulsive  $\Delta v$  versus transfer duration  $\Delta t$ . The evaluation will be based on the Pareto-optimal set of submitted solutions.

### 1. Mission Scenario

| ITEM                    | REQUIREMENT / VALUE                                                                          |
|-------------------------|----------------------------------------------------------------------------------------------|
| INITIAL ORBIT           | GTO with inclination 24.5°.                                                                  |
| TARGET ORBIT            | Circular geostationary orbit with inclination 0° at the simplified GEO-KOMPSAT-2A longitude. |
| INITIAL EPOCH           | 2026-06-01 00:00:00 UTC.                                                                     |
| TRANSFER DURATION LIMIT | 1 day $\leq \Delta t \leq$ 30 days.                                                          |
| DYNAMICS MODEL          | Two-body Earth-centered inertial propagation.                                                |

### 2. Given Constants and Coordinate Assumptions

| QUANTITY                      | SYMBOL                                   | VALUE                      | UNIT / NOTE                     |
|-------------------------------|------------------------------------------|----------------------------|---------------------------------|
| EARTH GRAVITATIONAL PARAMETER | $\mu_E$                                  | 398600.4418                | km <sup>3</sup> /s <sup>2</sup> |
| EARTH EQUATORIAL RADIUS       | $R_E$                                    | 6378.137                   | km                              |
| EARTH ROTATION RATE           | $\omega_E$                               | $7.2921150 \times 10^{-5}$ | rad/s                           |
| GEO RADIUS                    | $r_{GEO} = (\mu_E / \omega_E^2)^{(1/3)}$ | 42164.173                  | km                              |
| GEO ALTITUDE                  | $h_{GEO} = r_{GEO} - R_E$                | 35786.036                  | km                              |
| INITIAL EPOCH                 | $t_0$                                    | 2026-06-01 00:00:00        | UTC                             |
| JULIAN DATE AT T0             | $JD_0$                                   | 2461192.5                  | days                            |
| ERA AT T0                     | $\theta_{ERA,0}$                         | 249.1551677                | deg                             |
| GEO-KOMPSAT-2A LONGITUDE      | $\lambda_{GEO}$                          | 128.2                      | deg East                        |

### 3. Given Initial State and Target State

All vectors below are expressed in the Earth-centered inertial (ECI) frame at **t0 = 2026-06-01 00:00:00 UTC**.

#### 3.1 Initial GTO

| ELEMENT                        | VALUE        | UNIT / NOTE                          |
|--------------------------------|--------------|--------------------------------------|
| PERIGEE ALTITUDE               | 250          | km                                   |
| APOGEE RADIUS                  | 42164.173    | km                                   |
| SEMI-MAJOR AXIS                | 24396.155    | km                                   |
| ECCENTRICITY                   | 0.728312     | -                                    |
| INCLINATION                    | 24.5         | deg                                  |
| RAAN $\Omega$                  | 0            | deg                                  |
| ARGUMENT OF PERIAPSIS $\omega$ | 0            | deg                                  |
| TRUE ANOMALY $\theta$ AT T0    | 180          | deg; spacecraft starts at GTO apogee |
| <b>STATE COMPONENT</b>         | <b>Value</b> |                                      |
| R0,X [KM]                      | -42164.1729  |                                      |
| R0,Y [KM]                      | 0.0000       |                                      |
| R0,Z [KM]                      | 0.0000       |                                      |
| V0,X [KM/S]                    | 0.000000     |                                      |
| V0,Y [KM/S]                    | -1.458327    |                                      |
| V0,Z [KM/S]                    | -0.664598    |                                      |

#### 3.2 Target GEO at GEO-KOMPSAT-2A longitude

| STATE COMPONENT   | VALUE      |
|-------------------|------------|
| RGEO,X(T0) [KM]   | 41244.6079 |
| RGEO,Y(T0) [KM]   | 12577.3213 |
| RGEO,Z(T0) [KM]   | 0.0000     |
| VGEO,X(T0) [KM/S] | -0.917153  |
| VGEO,Y(T0) [KM/S] | 2.934683   |
| VGEO,Z(T0) [KM/S] | 0.000000   |

여기 있는 위성이 움직이는 타겟에 랑데뷰를 해야함.

### 4. Objective Function and Pareto Evaluation

The primary evaluation is multi-objective. Each valid solution is represented by the objective vector

$$F(x) = \left[ \sum \Delta v, \sum \Delta t \right].$$

A solution is Pareto optimal if no other valid solution has both lower or equal  $\Delta v_{\text{total}}$  and lower or equal  $\Delta t$ , with at least one objective strictly better. The instructor will compare submitted valid solutions using this Pareto-optimality concept.

### 5. Required Student Deliverables

| DELIVERABLE  | REQUIRED CONTENT                                                                                                                                    |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| FINAL REPORT | Problem formulation, assumptions, equations, optimization method, convergence discussion, final trajectory results, Pareto discussion               |
| MATLAB CODE  | All code used to generate the trajectory and figures. Code must run without external files except standard MATLAB functions.                        |
| ANIMATION    | 3D animation showing Earth, initial GTO, target GEO, spacecraft trajectory, maneuvers, and final GEO insertion.                                     |
| FIGURES      | At minimum: 3D trajectory, radius vs. time, $\Delta v$ maneuver locations, position error vs. time near final target, visibility intervals from KHU |
| RESULT TABLE | $\Delta t$ , $\Delta v_1$ , $\Delta v_2$ , ... , $\Delta v_2$ , $\Delta v_{\text{total}}$                                                           |

## 6. Animation Requirements

- Show the Earth as a sphere and use an equal-axis 3D view.
- Plot the initial GTO and the target GEO orbit before animating the optimized transfer.
- Mark the maneuver locations clearly.
- Animate the spacecraft motion using a uniform time grid so that the animation speed is not distorted by adaptive ODE step sizes.
- Show the target GEO spacecraft/location moving during the transfer, not fixed at its initial position.